

Cambridge (CIE) IGCSE Additional Maths



Your notes

Functions

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Language of Functions

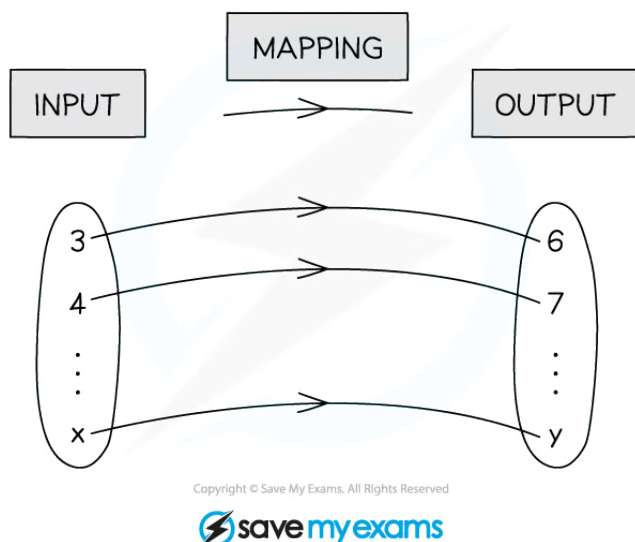


Your notes

Introduction to Functions

What is a mapping?

- A **mapping** takes an 'input' from one **set** of values to an 'output' in another



- Mappings can be
 - '**many-one**' (**many** 'input' values map to **one** 'output' value)
 - '**one-one**' (**one** 'input' value maps to **one** 'output' value)
 - You may also come across 'many-many' and 'one-many' functions

What is a function?

- A **function** is a mapping where every 'input' value maps to a **single** 'output'
- Therefore only **many-one** and **one-one** mappings are **functions**

What notation is used for functions?

- Functions are denoted by $f(x)$, $g(x)$, etc



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- e.g. $f(x) = x^2 - 3x + 2$
 - These would be pronounced as 'f of x', 'g of x', etc
- There is an alternative notation
 - e.g. $f: x \mapsto x^2 - 3x + 2$
 - Which may be pronounced 'the function f maps x to x-squared minus three x plus two'

How does a function work?

- A function has an **input** (x) and **output** ($f(x)$ or y)
- Whatever goes in the bracket (instead of x) with f , replaces the x on the other side
 - This is the input
- If the input is known, the output can be calculated
 - For example, given the function $f(x) = 2x + 1$
 - $f(3) = 2 \times 3 + 1 = 7$
 - $f(-4) = 2 \times (-4) + 1 = -7$
 - $f(a) = 2a + 1$
- If the output is known, an equation can be formed and solved to find the input
 - For example, given the function $f(x) = 2x + 1$
 - If $f(x) = 15$, the equation $2x + 1 = 15$ can be formed
 - Solving this equation gives an input of 7



Worked Example

A function is defined as $f(x) = 3x^2 - 2x + 1$.

a) Find $f(7)$.

The input is $x = 7$, so substitute 7 into the expression everywhere you see an x .



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$$f(7) = 3(7)^2 - 2(7) + 1$$

Calculate.

$$\begin{aligned} f(7) &= 3(49) - 14 + 1 \\ &= 147 - 14 + 1 \end{aligned}$$

$$f(7) = 134$$

b) Find $f(x + 3)$.

The input is $x = x + 3$ so substitute $x + 3$ into the expression everywhere you see an x .

$$f(x + 3) = 3(x + 3)^2 - 2(x + 3) + 1$$

Expand the brackets and simplify.

$$\begin{aligned} f(x + 3) &= 3(x^2 + 6x + 9) - 2(x + 3) + 1 \\ &= 3x^2 + 18x + 27 - 2x - 6 + 1 \\ &= 3x^2 + 16x + 22 \end{aligned}$$

$$f(x + 3) = 3x^2 + 16x + 22$$

A second function is defined $g : x \mapsto 3x - 4$.

c) Find the value of x for which $g : x \mapsto -16$.

Form an equation by setting the function equal to -16.

$$3x - 4 = -16$$

Solve the equation by first adding 4 to both sides, then dividing by 3.

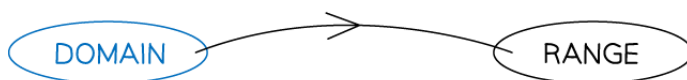
$$\begin{aligned} 3x - 4 &= -16 \\ 3x &= -12 \\ x &= -\frac{12}{3} \end{aligned}$$

$$x = -4$$

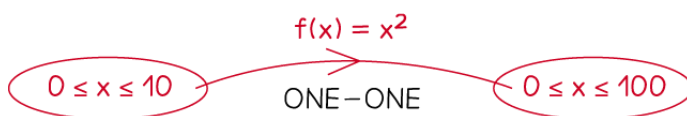
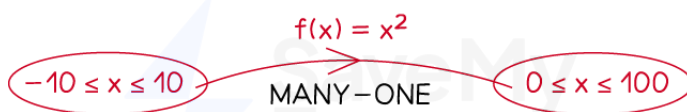
Domain & Range

What is the domain of a function?

- The **domain** of a function is the set of values that are allowed to be the 'input'
- A function is only fully defined once its domain has been stated
 - If a domain is not stated then it is assumed that the domain is the **largest set of possible values**
 - e.g. the largest set of possible values for the function $f(x) = \sqrt{x}$ would be $x \geq 0$
- Restrictions** on a domain can turn **many-one** functions into **one-one** functions



RESTRICTING THE DOMAIN



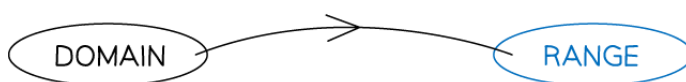
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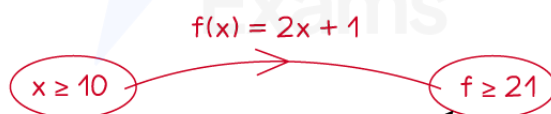
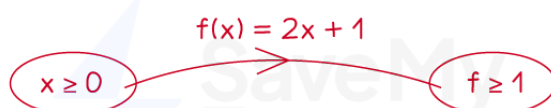
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What is the range of a function?

- The **range** of a function is the set of values of all possible 'outputs'
- The type of values in the range depend on the **domain**



THE RANGE IS AFFECTED BY THE DOMAIN



THE CHANGE IN THE DOMAIN HAS CHANGED THE RANGE

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How do I find a range from a given domain?

- The **domain** of a function is the set of values that are used as **inputs**
- The **range** of a function is the set of values that are given as **outputs**
- Finding the **range** of a function involves determining all possible **output** values from a given **domain**
 - This may need to be done by calculating each output value individually by **applying the function** to each input value
 - Or by considering the **shape** or **pattern** of the function
- To graph a function we use the **inputs as the x-coordinates** and the **outputs as the y-coordinates**
 - $f(2) = 5$ corresponds to the coordinates (2, 5)



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- Graphing the function can help you visualise the range
 - For example the range of the function $f(x) = x^2$ for a domain of all real values of x will be $f(x) \geq 0$ as the y -coordinates on the graph are all greater than or equal to zero



Worked Example

The many-one function, $f(x)$, is given by

$$f(x) = (x - 3)^2$$

for all values of x .

a) State the range of $f(x)$.

The 'output' from the function f is a squared value and so will be positive, or zero.

$$f \geq 0$$

b) The domain of $f(x)$ is changed to $x > 5$. Write down the changed range of $f(x)$.

$$\text{As } x > 5, f(x) > (5 - 3)^2$$

$$\therefore f > 4$$

The Modulus Function

What is the modulus function?

- The **modulus function** makes **any** 'input' **positive**
 - This is sometimes called the **absolute** value (of the input)
 - The modulus function is indicated by a pair of vertical lines being written around the input
 - Similar to how brackets are used
 - e.g. $|7| = 7$, $|-7| = 7$

What is the relationship between a function and its modulus?

- For an 'output' such that $f(x) \geq 0$, then $|f(x)| = f(x)$
 - Both the function and its modulus are positive

- For an 'output' such that $f(x) < 0$, then $|f(x)| = -f(x)$
 - The function value is **negative**, but its modulus is **positive**



Your notes



Your notes

Inverse Functions

Inverse Functions

What is an inverse function?

- An **inverse function** does the exact opposite of the function it came from
 - For example, if the function “doubles the number and adds 1” then its inverse is
 - “subtract 1 and halve the result”
- It is the **inverse** operations in the **reverse** order

How do I write inverse functions?

- An inverse function f^{-1} can be written as $f^{-1}(x) = \dots$ or $f^{-1}: x \mapsto \dots$
 - For example, if $f(x) = 2x + 1$ its inverse can be written as
 - $f^{-1}(x) = \frac{(x-1)}{2}$ or $f^{-1}: x \mapsto \frac{(x-1)}{2}$

How do I find an inverse function?

- The easiest way to find an inverse function is to 'cheat' and swap the x and y variables
 - Note that this is a useful method but you **MUST** remember **not** to do this in any other circumstances in maths
- STEP 1** Write the function in the form $y = \dots$ e.g. $y = 2x + 1$
- STEP 2** Swap the x 's and y 's to get $x = \dots$ e.g. $x = 2y + 1$
- STEP 3 Rearrange** the expression to make y the subject again
$$x - 1 = 2y$$
$$\frac{x-1}{2} = y \rightarrow y = \frac{x-1}{2}$$
- STEP 4** Rewrite using the correct **notation** for an inverse function
 - either as $f^{-1}(x) = \dots$ or $f^{-1}: x \mapsto \dots$
 - y should not exist in the final answer

- e.g. $f^{-1}(x) = \frac{x-1}{2}$



Your notes

How does a function relate to its inverse?

- If $f(3) = 10$ then the input of 3 gives an output of 10
 - The inverse function undoes $f(x)$
 - An input of 10 into the inverse function gives an output of 3
 - If $f(3) = 10$ then $f^{-1}(10) = 3$
- $ff^{-1}(x) = f^{-1}f(x) = x$
 - If you apply a function to x , then immediately apply its inverse function, you get x
 - Whatever happened to x gets undone
 - f and f^{-1} cancel each other out when applied together
- If $f(x) = 2^x$ and you want to solve $f^{-1}(x) = 5$
 - Finding the inverse function $f^{-1}(x)$ in this case is tricky (impossible if you haven't studied logarithms)
 - instead, take f of both sides and use that ff^{-1} cancel each other out:

$$\begin{aligned} ff^{-1}(x) &= f(5) \\ x &= f(5) \\ x &= 2^5 = 32 \end{aligned}$$



Worked Example

Find the inverse of the function $f(x) = 5 - 3x$.

Write the function in the form $y = 5 - 3x$ and then swap the x and y .

$$\begin{aligned} y &= 5 - 3x \\ x &= 5 - 3y \end{aligned}$$

Rearrange the expression to make y the subject again.

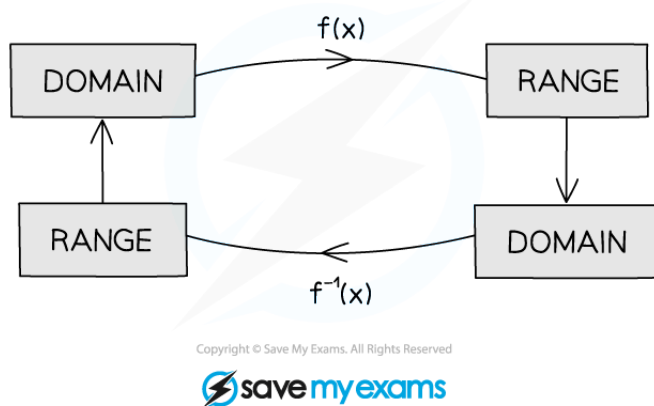
$$\begin{aligned}x &= 5 - 3y \\x + 3y &= 5 \\3y &= 5 - x \\y &= \frac{5 - x}{3}\end{aligned}$$

Rewrite using the correct **notation** for an inverse function.

$$f^{-1}(x) = \frac{5 - x}{3}$$

Domain & Range of Inverse Functions

How do I find the domain and range of inverse functions?



- The **range** of a function will be the **domain** of its **inverse** function
- The **domain** of a function will be the **range** of its **inverse** function



Worked Example

A function is defined as $f(x) = \sqrt{3x - 2}$, $x > \frac{3}{2}$.



Your notes

Write down the domain and range of $f^{-1}(x)$.

The domain of an inverse function is the range of the function.

The range of $f(x)$ is

$$f > 0$$

$\therefore$ The domain of $f^{-1}(x)$ is $x > 0$

The range of an inverse function is the domain of the function.

$\therefore$ The range of $f^{-1}(x)$ is $f^{-1} > \frac{3}{2}$

Graphs of Inverse Functions

How are the graphs of a function and its inverse related?

- The graph of an **inverse function**, $y = f^{-1}(x)$, is a **reflection** of the graph of the function, $y = f(x)$, in the line $y = x$
- Key features of the graph of $y = f(x)$ will be reflected, such as
 - x and y axes **intercepts**
 - turning points**
 - asymptotes**

How do I sketch the graph of an inverse function?

- STEP 1**
 - Sketch** the line $y = x$, and if need be, the graph of $y = f(x)$
- STEP 2**
 - Reflect** the graph of $y = f(x)$ in the line $y = x$
 - Remember it is a **sketch**, but the graphs together should look like reflections
 - Consider points where the reflected graph will **intersect** the x and y axes
 - e.g. The point $(4, 0)$ will be reflected to the point $(0, 4)$



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- Consider any **asymptotes** on the graph of $y = f(x)$ - these will also be need reflecting
 - e.g. The asymptote (line) $x = -2$ will be reflected to the line $y = -2$
- Consider any restrictions on the **domain** and **range** of $f(x)$
 - e.g. If the domain is $x > 0$ only sketch the graph for positive values of x
- **STEP 3**
 - Label key points on the sketch of $y = f^{-1}(x)$ and state the equations of any asymptotes
- This process works the other way round - the graph of $y = f(x)$ can be sketched from the graph of $y = f^{-1}(x)$



Examiner Tips and Tricks

- If **not** given, sketch the graphs of $y = f(x)$ and $y = x$ to help sketch the graph of the inverse, $y = f^{-1}(x)$
- If the graph of $y = f(x)$ is given you do not need to know the expression for $f(x)$ to sketch $y = f^{-1}(x)$
 - Just reflect whatever is given in the line $y = x$

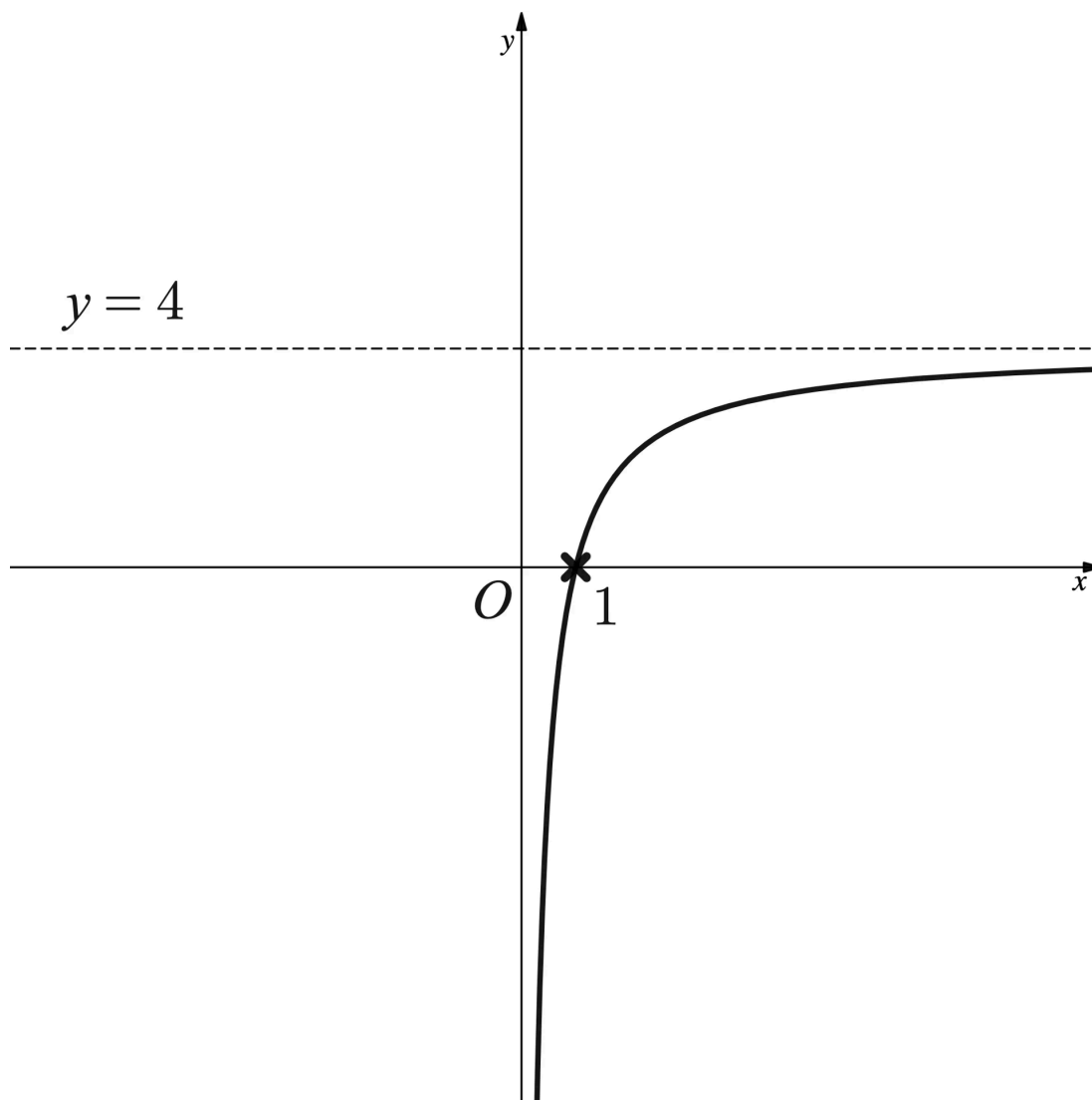


Worked Example

The diagram below shows the graph of $y = f(x)$, where $f(x) = 4 - \frac{4}{x}$, $x > 0$.



Your notes



a) On a copy of the diagram, sketch the graph of $y = f^{-1}(x)$. Label the point where the graph crosses the y -axis and write down the equation of the asymptote.

The graph of an inverse function is the reflection of the graph of that function in the line $y = x$.

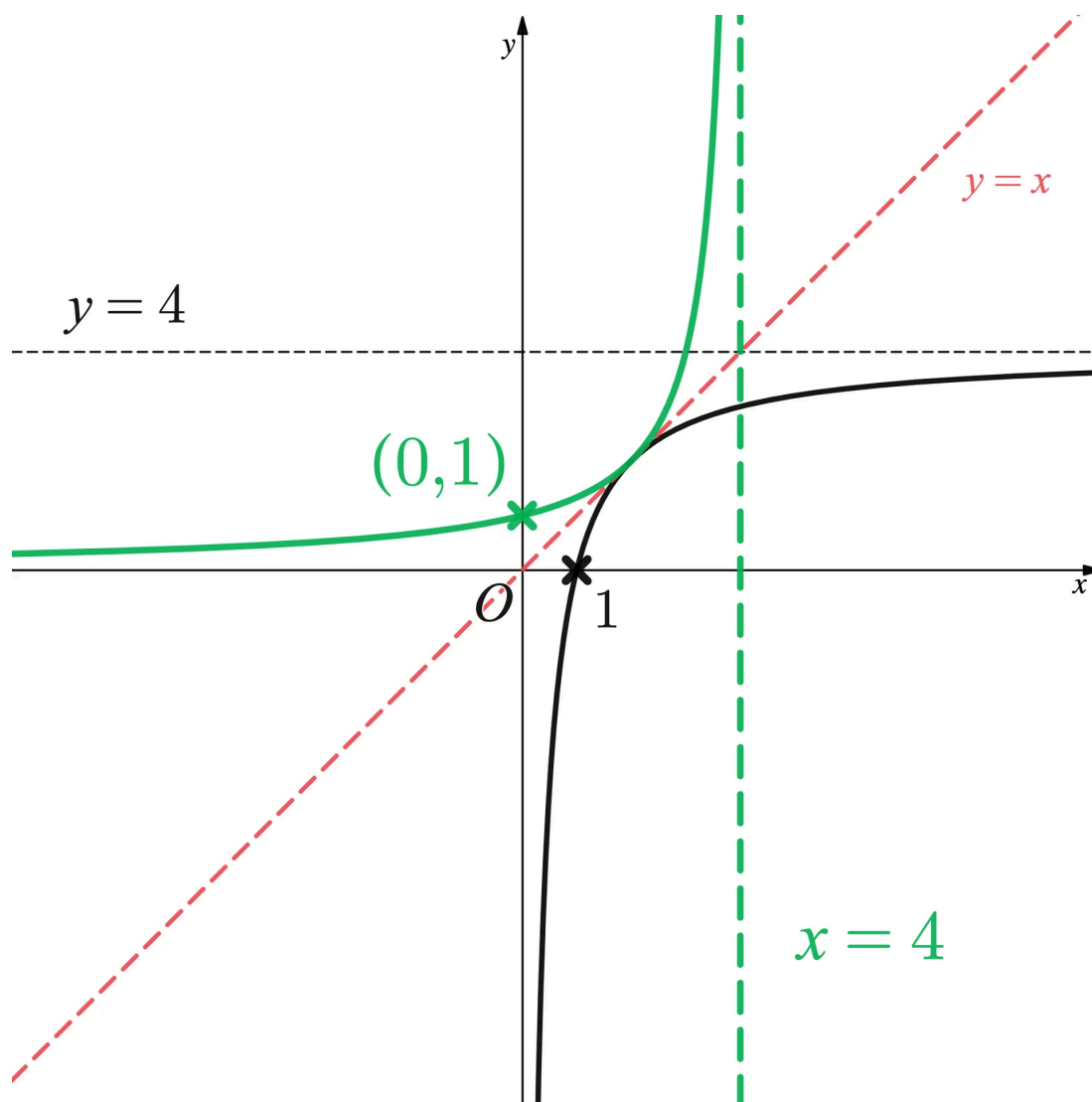
Draw the line $y = x$ to help sketch the inverse function.

The x -axis intercept $(1, 0)$ becomes the y -axis intercept, $(0, 1)$.

The (horizontal) asymptote $y = 4$ will become the (vertical) asymptote $x = 4$.



Your notes



b) Use your sketch, or otherwise, to write down the value of x such that $f(x) = f^{-1}(x)$.

This will be the point at which the two graphs meet.

The point will be on the line $y = x$ so there is no need to work out $f^{-1}(x)$.

By sketching the graph in part (a) this point (with coordinates $(2, 2)$) should have already been considered. Only the x value is required.

$$x = 2$$

The x value could also be found by solving $f(x) = x$.



Your notes



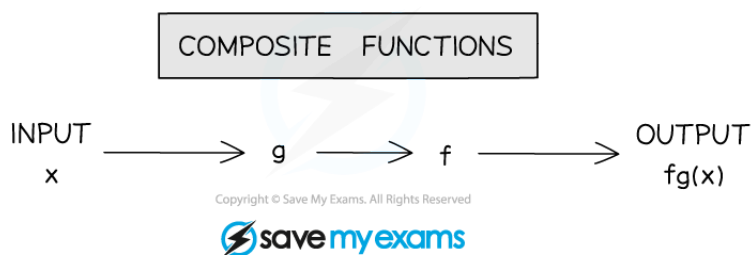
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Composite Functions

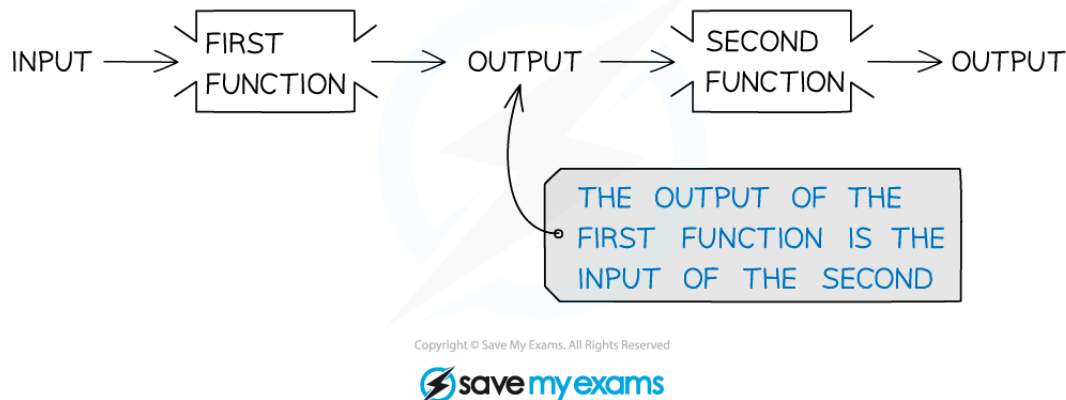
Composite Functions

What is a composite function?

- A **composite function** is where one function is applied after another function



- The '**output**' of one function will be the '**input**' of the next one
- Sometimes called **function-of-a-function**
- A composite function can be denoted
 - $fg(x)$
 - $f(g(x))$
 - All of these mean "f of g(x)"





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How do I work with composite functions?



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- **Recognise** the notation
 - **$fg(x)$** means “f of g of x”

The order matters

- First apply **g** to **x** to get **$g(x)$**
- Then apply **f** to **$g(x)$** to get **$f(g(x))$**
- Always start with the function **closest to the variable**
- **$fg(x)$** is not usually equal to **$gf(x)$**

Special cases

- **$fg(x)$** and **$gf(x)$** are generally different but can sometimes be the same
- **$ff(x)$** is written as **$f^2(x)$**
 - Note that **trig functions** are **exceptions** to this rule
 - e.g. **$\sin^2(x)$** means **$(\sin(x))^2$** not **$\sin(\sin(x))$**
- For inverse functions, **$ff^{-1}(x) = f^{-1}f(x) = x$**



Your notes

$$f(x) = x^2 \quad g(x) = 2x$$

$$fg(x) = f(2x) = (2x)^2 = 4x^2$$

$$gf(x) = g(x^2) = 2(x^2) = 2x^2$$

$$fg \neq gf$$

$$f(x) = 3x - 2 \quad g(x) = 6x - 5$$

$$fg(x) = f(6x - 5) = 3(6x - 5) - 2 = 18x - 17$$

$$gf(x) = g(3x - 2) = 6(3x - 2) - 5 = 18x - 17$$

$$fg = gf$$

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Worked Example

Two functions, $f(x)$ and $g(x)$ are

$$f(x) = x^2 + 3x - 2$$

$$g(x) = x + 3$$

a) Find $f(3)$ and $g(3)$.

$$f(3) = (3)^2 + 3(3) - 2$$

$$= 9 + 6 - 2$$

$$g(3) = 3 + 3$$

$$f(3) = 13, g(3) = 6$$

b) Find, in terms of x , $fg(x)$.

g is the first function to be applied ...



Your notes

$$\begin{aligned}\therefore fg(x) &= f[x+3] \\ &= (x+3)^2 + 3(x+3) - 2 \\ &= x^2 + 6x + 9 + 3x + 9 - 2 \\ fg(x) &= x^2 + 9x + 16\end{aligned}$$

Domain & Range of Composite Functions

How do I find the domain and range of composite functions?

- Use logic to determine the domain and range of a **composite function**
- For $fg(x)$ the first function to be applied will be g
 - So, at best, the **domain** of $fg(x)$ will be the **same** as the **domain** of $g(x)$
- However, for this to be the case, the range of $g(x)$ must be contained within the domain of $f(x)$
 - If this is *not* the case, then **restrictions** on the **domain** of $fg(x)$ will be required
- Similarly, at best, the **range** of $fg(x)$ will be the same as the **range** of $f(x)$
 - But if the domain of $f(x)$ has been affected, the range of $fg(x)$ will also be affected



Examiner Tips and Tricks

- **Domain** and **range** are important in composite functions like $fg(x)$
 - the 'output' (range) of g **must** be in the domain of $f(x)$, so $fg(x)$ could exist, but $gf(x)$ may not (or not for some values of x)



Worked Example

Two functions, $f(x)$ and $g(x)$ are defined as follows

$$f(x) = \frac{1}{x}, \quad 0 < x < 1$$



Your notes

$$g(x) = x^2, \quad x > 1$$

a) Write down the range of $f(x)$ and the range of $g(x)$.

As the domain of $f(x)$ is $0 < x < 1$, $\frac{1}{x}$ will always be greater than 1,

The range of $f(x)$ is $f > 1$

The square of any value will be positive or zero, but here, $x = 0$ is not included in the domain for $g(x)$.

The range of $g(x)$ is $g > 1$

b) Use your answers to (a) to help explain why $fg(x)$ does not exist.

g is the first function to be applied The range of g would need to be contained within the domain of f But the range of g is $g > 1$ which is outside the domain of f which is $0 < x < 1 \therefore fg(x)$ does not exist

c) Find the range of $gf(x)$,

f is the first function. The range of f is $f > 1$. This is the same as the domain of g ($x > 1$).

The range of gf is $gf > 1$